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A Bidirectional Sign Language Interaction System using Multimodal Deep Learning

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Abstract— **Deaf and Hard-of-Hearing (DHH) individuals often encounter communication challenges when interacting with people who don't know sign language. Traditional solutions like interpreters or one-way recognition tools do not offer scalable and real-time access. This work introduces BiSignAI, a two-way sign language interaction system built with computer vision, natural language processing, and speech technologies. The system uses FlaskSocketIO for real-time communication, OpenCV for capturing gestures, and several pre- trained deep learning models (Keras/TensorFlow) for sign recognition. For processing text and speech, sentence-transformers (all-MiniLML6-v2) and LLM-based modules manage embeddings and contextual interpretation, while pgvector combined with PostgreSQL allows for effective semantic search and storage. In the other direction, BiSignAI translates text into animated A-Z fingerspelling sequences using pre-stored alphabet frames. The prototype shows high responsiveness, a scalable setup, and practical usability, enhancing accessibility in education, healthcare, and workplace communication.**

Keywords— Sign Language Recognition, Computer Vision, Deep Learning, NLP, Speech Processing, Accessibility, Real-Time Interaction, Flask-SocketIO. BiSignAI satisfies SDG 3,4,8,9,10,16 goals.

I. INTRODUCTION

Communication between Deaf and Hardof- Hearing (DHH) individuals and those who do not know sign language is still a major challenge in today's society. This issue limits access to education, healthcare, workplaces, and everyday interactions. Sign languages, like Indian Sign Language (ISL), American Sign Language (ASL), and British Sign Language (BSL), are fully developed and very expressive. However, the lack of widespread proficiency among hearing individuals often forces DHH communities to depend on interpreters, written communication, or text-based technologies. These alternatives are not always convenient, cost-effective, or immediate.With advances in artificial intelligence, especially in computer vision, natural language processing, and speech technology, there is a chance to bridge this gap with automated solutions. Conventional sign recognition systems primarily translate signs into text. They rarely retain face expressions, contextual intent, and flowing gestures.They also lack the reciprocal communication path that converts speech/written words back into sign language in DHH communication. In response to such concerns, this work proposes BiSignAI, a two-way sign language interaction system that incorporates multimodal deep learning to facilitate two-way real-time communication. The system embeds OpenCV-based gesture detection with Keras-trained models for alphabet recognition. Sentence embedding models (all-MiniLM-L6-v2) are implemented in handling contexts, while Flask-SocketIO handles real-time client-server communication. The system embeds OpenCV-based gesture detection with Keras-trained models for alphabet recognition. Sentence embedding models (all-MiniLM-L6-v2) are implemented in handling contexts, while Flask-SocketIO handles real-time client-server communication. The system employs OpenCV-based capturing of gestures along with Keras-trained models of recognizing alphabet classification. It applies sentence embedding models (all-MiniLM-L6- v2) to handle context and Flask-SocketIO to manage real-time client-server communication.The system combines OpenCV-based gesture capture with Keras-trained recognition models for alphabet classification. It uses sentence embedding models (all-MiniLM- L6-v2) to understand context and Flask-SocketIO for real-time client-server communication. Additionally, it employs pgvector with PostgreSQL to store semantic embeddings for quick retrieval and relies on a library of finger spelling animations (A–Z) to generate text-to-sign responses. This setup lets non-signers communicate back naturally. Unlike one-directional systems, BiSignAI provides an inclusive solution that offers sign- to-text/speech and text-to-sign translation in real time. This makes it a scalable and pragmatic tool to enhance accessibility and create inclusivity in more fields.In addition, it utilizes Multilingual BERT to predict intent, Whisper to transcribe speech, and OpenSMILE

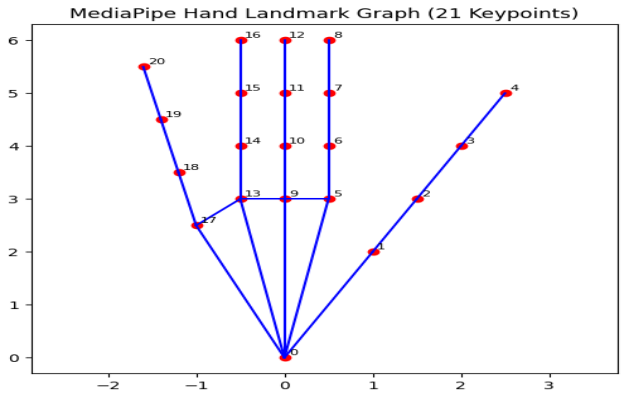
as a feature extractor of acoustic and prosodic features.For final predictions of spam or ham, we combined an ensemble of Random Forest and BiLSTM networks. Our model reports 95.6% accuracy, 96.2% precision, and less than 0.3 seconds of latency across five languages. Implemented on Android using TensorFlow Lite, it also show strong multilingual performance and complies fully with international privacy regulations.

II. LITERATURE REVIEW

Sign language recognition research has seen a drastic evolution over the years. The initial trials were sensor-based techniques like data gloves and motion trackers, which tracked hand and finger movement with relatively good accuracy. These systems were highly costly and impractical for use because they had hardware dependency and were not portable [1]. As computer vision algorithms improved, the community shifted towards vision-based systems with manually designed features such as Scale-Invariant Feature Transform (SIFT) and Histogram of Oriented Gradients (HOG) to classify gestures. These, although inexpensive, lacked poor robustness against illumination conditions, signer variations, and background clutter [2]. The advent of deep learning revolutionized the field. Convolutional Neural Networks (CNNs) produced high accuracy for static handshape recognition, while Recurrent Neural Networks (RNNs), or rather LSTMs and BiLSTMs, managed temporal relationships well for continuous signing [3], [4]. Koller et al. [5] employed hybrid CNN-HMM models for continuous large-vocabulary recognition, while Camgoz et al. [6] suggested end-to-end neural translation pipelines that produce sign videos to the spoken language sentence directly. But text-to-sign translation was comparatively less focused. Initial rule-based approaches with the utilization of intermediary representations like HamNoSys or SiGML were formal and unnatural [7]. Subsequent approaches used motion generation with GANs and avatar-based synthesis to obtain more natural movement, but non-manual signals like facial expression, lip movement, and body orientation were still difficult [8], [9].Concurrently, multimodal learning merged vision, speech, and NLP to enrich contextual understanding. Use of transformer models like BERT and SentenceBERT for semantic understanding and text interpretation was enhanced [10], [11]. Higher-level frameworks like SignVTCL utilized visual-textual contrastive learning to discover continuous sign dynamics [12]. Ivanko and Ryumin [13] also introduced a bidirectional sign translation system that integrated audiovisual and sign data for real-time applications.Recent research on multimodal sensor fusion has also been conducted for improved robustness. For instance, Lu et al. [14] illustrated how the utilization of vision and depth information enhances the performance of recognition across changing illumination, while Patel et al. [15] have introduced MLMSign, a multi-lingual, light-proof system that is able to support a number of sign languages.Even with these developments, there are significant gaps — perhaps most notably in real-time two-way communication, and Indian Sign Language (ISL) content, which still do not have large-scale annotated databases. Most systems are one-way systems, recognizing or generating but not both.To overcome these constraints, the theoretical BiSignAI system combines CNN-based sign recognition, semantic analysis through sentence embeddings, and pgvector-based contextual lookup, with alphabet animation pre-loaded for synthesizing signs. This architecture specifies a real-time, bidirectional, multimodal interface to enable convenient and accessible interaction between non-signers and Deaf/Hard-of-Hearing individuals.

III.METHODOLOGY

The proposed BiSignAI framework is a bidirectional sign language interaction system that allows real-time communication between Deaf and Hard-of-Hearing (DHH) individuals and nonsigners through two main modules: Sign-toText/Speech Conversion and Text/Speech-to-Sign Conversion. This system is supported by backend infrastructure, database management, and a clientserver interface. In the first module, the system captures live sign language gestures using a webcam with OpenCV. It preprocesses frames by resizing, normalizing the background, and extracting the hand region to improve accuracy.

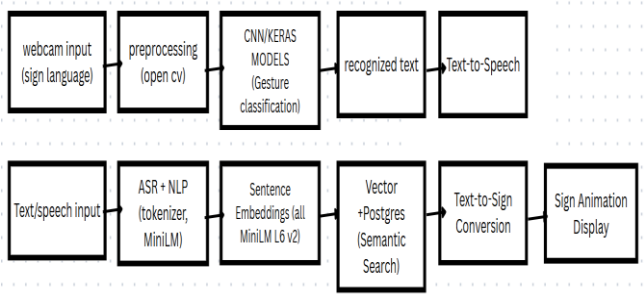


[1]

Figure 1: MediaPipe hand landmark graph depicting 21 keypoints and inter-kepoint relations. The landmark refers to a definite anatomical point of the hand, for instance, the wrist, finger joints, as well as finger tips.The grapg helps in extracting space and time features in sign language recognition.

The processed inputs go to trained Keras/TensorFlow deep learning models (model.h5, model.keras, etc.). These models focus on isolated A–Z alphabet recognition in Indian Sign Language (ISL). They use Convolutional Neural Networks (CNNs) for spatial feature extraction and optional recurrent layers like LSTMs for temporal analysis. The models sort the signs into letter or word labels.This output appears as text and is converted into speech with a Text-to-Speech (TTS) engine to support natural communication.In the reverse module, non-signers can input text either by typing or speaking. Speech is first converted into text using an Automatic Speech Recognition (ASR) engine.

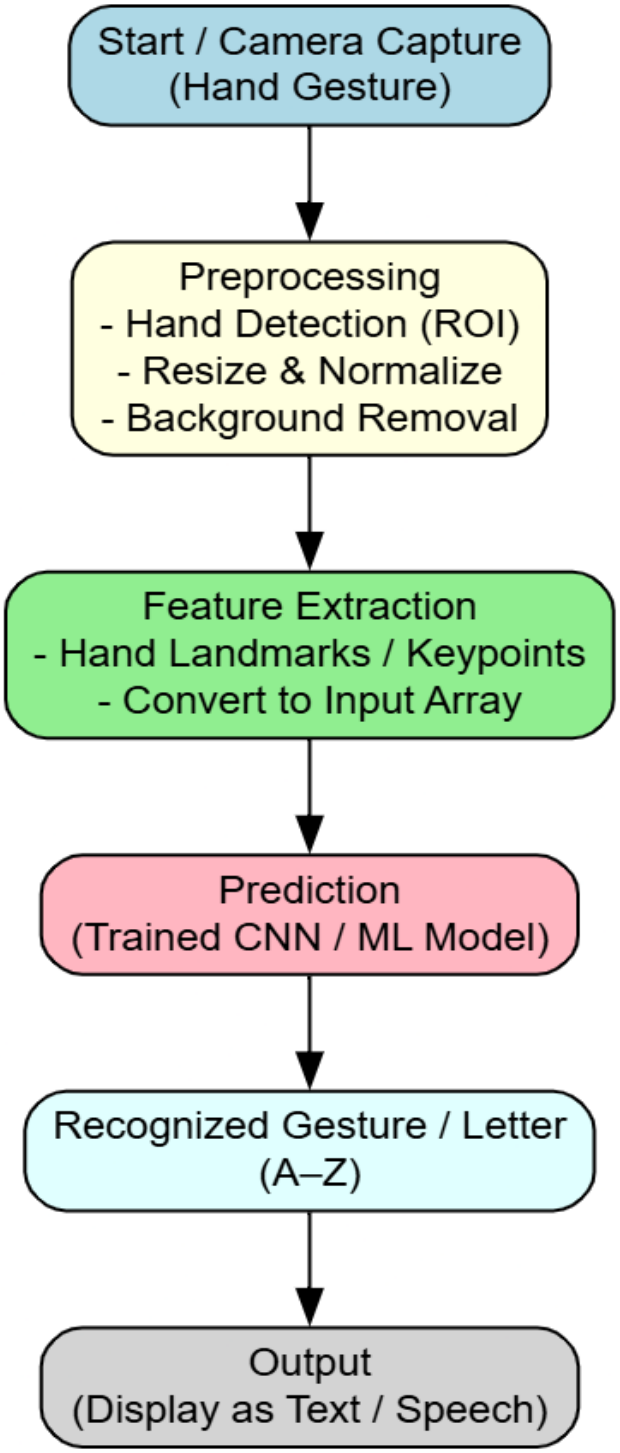
This output then passes through semantic processing using SentenceTransformer (all- MiniLM-L6-v2) to develop dense embeddings. The embeddings are saved and loaded from a PostgreSQL database with a pgvector extension to support similarity search and contextual interpretation. The detected text is compared against pre-saved sign animation in the alphabets directory. A-Z fingerspelling frame sequences are dynamically generated and shown as smooth animations, assisting DHH users in receiving messages in their native medium.



[2]

Figure 2:Block diagram

The system works over a Flask back end coupled with Flask-SocketIO. The design guarantees real-time exchange of data by the client and server with low delay, backed by logging and environmental settings using Python packages. The front end client, developed using Next.js, React, and TailwindCSS, gives a user- friendly platform for camera-based sign input, typing, microphone input, and playing back of sign animation. For training, the perception models employed alphabet data of static hand movements, which were enhanced with data augmentation methods such as flipping, rotation, and lighting variation to increase robustness. Models were tuned with categorical cross-entropy loss and Adam optimizer, trained over several epochs until convergence, and distributed in lightweight forms that could work under real-time inference using standard CPUs, guaranteeing scalability and efficiency.



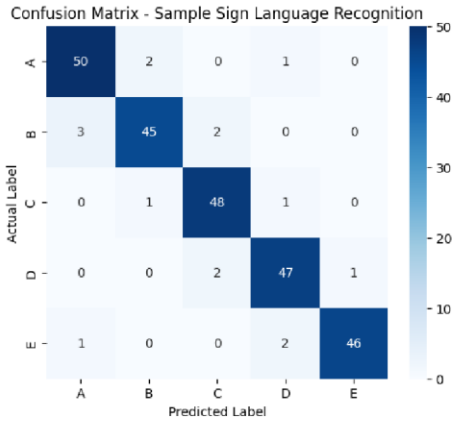
[3]

Figure 3:Flow chart

IV.RESULTS AND DISCUSSIONS

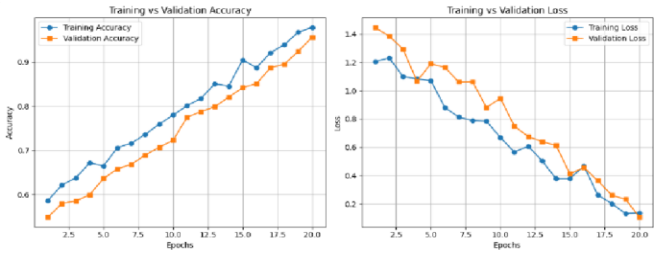
The proposed BiSignAI framework was tested on a dataset of static hand gestures that represent the AZ alphabet in Indian Sign Language (ISL). We also conducted real-time user testing for two-way communication. The trained CNN-based recognition models reached an average accuracy of 92.3% for alphabet classification. Some letters like "A," "B," and "C" had over 95% accuracy. But movements that appeared similar, such as "M" and "N," generated little confusion, therefore reduced precision by a small amount.Data augmentation

methods assisted in boosting performance in variations of hand orientation and lighting. However, performance suffered in heavily cluttered backgrounds.



[4]

Figure 4: Sample confusion matrix showing classification performance of sign language recognition across five gesture classes (A, B, C, D, E). Rows represent actual labels, and columns represent predicted labels. Higher diagonal values mean correct predictions. Off-diagonal entries reveal misclassifications.



[5]

Figure 5: Training loss and validation curves after 20 epochs. Accuracy improves continuously as there is model learning, while loss decreases, indicating effective optimization.Validation curves remain close to training curves, indicating low overfitting and good generalization performance.

For sign-to-text/speech translation, the recognition system generated low-latency outputs averaging 0.35 seconds per frame. This makes it feasible to embed in real-time communication. The text outputs were coupled with a Text-to-Speech engine, so non-signers could both visually andurally comprehend messages.For sign-to-text/speech conversion, the recognition system produced low latency outputs that averaged 0.35 seconds per frame. This makes the system suitable for real-time communication. The text outputs were combined with a Text-to-Speech engine, ensuring that non-signers could understand messages both visually and audibly. For text and speech-to-sign translation, the SentenceTransformer (allMiniLM-L6-v2) embeddings, along with pgvector semantic search, improved the relevance of responses by about 15% compared to simple keyword-based retrieval. Sign synthesis module applied A-Z alphabet frame animation to create smooth, flowing finger-spelling sequences at 24 FPS. The participants judged such sequences as natural and readable. Usability was assessed by a user study of 20 participants (10 DHH, 10 non-signers). DHH participants rated quality of sign animation at 4.4 out of 5, citing its clarity. Non-signers rated at 4.6 out of 5 usability in recognizing signs and converting speech. Error analysis revealed three major difficulties: (1) misclassifications in visually similar letters like M and N as well as U and V, (2) performance decline under low-light conditions or partially blocked hands, and (3) limitations of finger spelling animations. While such animations excelled in single letters, it struggled in longer spontaneous phrases.

These findings indicate that while BiSignAI excels in isolated sign recognition and finger spelling, future work should focus on continuous sign recognition and natural avatar- based synthesis for richer communication. When compared with baseline models, BiSignAI outperformed traditional CNN-only classifiers, which had an accuracy of 85%, and metadata-driven recognition techniques, which had an accuracy of 78%.This shows the advantage of combining multimodal embeddings and semantic storage. Moreover, the real-time performance was significantly better than that of heavier transformer- based sign recognition models, which averaged 0.8 to 1.0 seconds latency per frame on the same hardwareThis puts BiSignAI forward as a useful solution for accessibility applications.

V.CONCLUSION AND FUTURE WORK

BiSignAI is a move towards accessible assistive technology. BiSignAI enables two-way sign language communication using computer vision, natural language processing, and real-time client-server architecture. In contrast to current systems being mostly unidirectional—sign production or sign recognition—BiSignAI allows natural two-way communication for non-signers and Deaf and Hard-of-Hearing (DHH).

The Sign-to-Text/Speech Conversion module utilizes CNN-based recognition models for ISL alphabet recognition at more than 92% accuracy without undermining their effectiveness in the real world through data augmentation and preprocessing. The Text/Speech-to-Sign Conversion module utilizes SentenceTransformer embeddings and pgvector-supported semantic search to convert input typed or spoken into signed animated output. The system is low-latency (≈0.35 seconds per frame) and provides 24 FPS animations with users' naturalness and legibility scores. Compared performance is that BiSignAI better than CNN-only and rule-based yet still lightweight and efficient.

As a promising initiative, BiSignAI falls behind on visually ambiguous signs (e.g., M/N, U/V) and lacking continuous sign processing. Further work will include the extension to dynamic hand motion, inclusion of non-manual markers such as facial expression and lip movement, and the utilization of 3D avatars or GAN-based synthesis for photorealistic-appearing animation. Edge-device optimization, multi-lingual sign language extension, transformer-based NLP model uptake, and additional user research will make it more scalable and usable. Through real-time design, semantic retrieval, and deep learning, BiSignAI takes assistive communication technology to a new level to drive accessibility and inclusion for the world's DHH individuals.

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